

Victor Ndunda

AI Engineer · Founder · **Production systems, not demos**

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PRODUCTION AI SYSTEMS SHIPPED

11K+

LEARNERS ON A SHIPPED CLIENT PRODUCT

22+

DATA SOURCES INTEGRATED

91%

ON-DEVICE CV ACCURACY

01 Profile

AI engineer and founder building **production multi-agent intelligence systems**, agricultural AI, and open-source developer tools for African and global markets. Founder of **Busara AI** — a multi-agent data intelligence platform running a 7-stage parallel DAG pipeline with real statistical algorithms (Holt-Winters, OLS, K-Means++, Granger causality, GARCH). Building **KilimoPRO**, an offline-first app for 8 IGAD countries with on-device crop disease detection (TensorFlow Lite) and market price forecasting (GARCH/VaR) over 22+ data sources. Creator of **AgentReplay** — deterministic replay and counterfactual debugging for AI agents (99 tests, 0.67% overhead, 100% reproduction fidelity, MIT). Builder of **Verxlite** (universal AI workflow agent) and a **Go-based real-time fraud detection engine** (7 statistical detectors, ~33 μ s/score, 84% recall). 5+ years shipping Go, TypeScript, Python, Flutter, and ML systems that serve real users — in English, Français, and Kiswahili, on low-end phones.

Founder & Lead AI Engineer

2023 — PRESENT

[Busara AI](#) · busaraai.com

- Designed and shipped a **multi-agent platform** with a 7-stage parallel DAG pipeline (Ingest › Engineer › Detect › Forecast › Infer › Report), circuit breakers and smart caching — serving live users.
- Implemented real statistical algorithms from scratch: Holt-Winters forecasting, Z-score/IQR/EWMA anomaly ensembles, OLS regression, Granger causality, K-Means++ clustering.
- Integrated GLM-4.6, GPT, Claude and Gemini into a multi-LLM council with streaming responses and citation-grounded RAG; cut average analysis latency ~40% via caching and parallel DAG execution.

TypeScript

Next.js 15

React 19

PostgreSQL

Redis

Docker

Kubernetes

PWA

Creator & Maintainer — AgentReplay

2026 — PRESENT

[Open Source](#) · github.com/gadda00/agentreplay

- Shipped an open-source **deterministic replay and counterfactual debugging layer** for AI agents — records every LLM call, tool response, clock read and RNG draw, then replays the run bit-exact with **zero model calls**.
- Measured **0.67% recording overhead** (vs 5% target), **100% reproduction fidelity**, **99 tests** passing; first-class LangGraph integration plus OpenAI, Anthropic, CrewAI and AutoGen adapters.
- Counterfactual mutation engine turns “would the agent still have done X if Y differed?” into a 5-second, zero-cost experiment; captured failures become permanent CI regression tests.
- Published a 22-minute research whitepaper covering METR time-horizon research and the observability gap.

Python

LangGraph

OpenAI SDK

Anthropic SDK

Click CLI

GitHub Actions

Founder & Lead Engineer — KilimoPRO

2022 — PRESENT

[Open Source](#) · [kilimo.pro](#)

- Built an **offline-first Flutter app** for African farmers with on-device crop disease detection via TensorFlow Lite — runs on \$80 Android phones, syncing when connectivity returns.
- Engineered market price forecasting (GARCH/EWMA/VaR) and a multi-LLM advisory council covering agronomy, markets, climate and finance; integrated **22+ verified data sources** (KilimoSTAT, FAOSTAT, Earth Engine, KALRO).
- Aligned with Kenya's NASIP 2026-2030 strategy; SMS/USSD channels reach feature-phone users across 8 IGAD countries.

Flutter

TensorFlow Lite

MobileNetV3

Node.js

Africa's Talking SMS/USSD

AI Consultant & Founder

2022 — PRESENT

[Independent](#) · Nairobi · Remote worldwide

- End-to-end AI delivery: scoping audit > architecture > build > production deployment > team training, for clients across fintech, agriculture, logistics and SMBs.
- Built **IntelliFlow** — a 33-agent TypeScript system with self-evolving trajectory-based learning that became the foundation of Busara AI.
- Runs a fully first-party commercial pipeline: instant estimate assistant, scheduler, proposals, contracts, invoicing and payments.

Data Scientist & Analytics Engineer

2019 — 2022

[Independent](#) · Nairobi · Remote

- Delivered financial forecasting systems (ARIMA, Holt-Winters, exponential smoothing, backtesting) — **87.3% mean accuracy** across 12 financial metrics.
- Built a real-time fraud-detection microservice in Go — Z-Score + IQR + Velocity detectors in a weighted-vote ensemble, scoring transactions in ~50–300 μ s for 50,000+ TPS.
- Shipped customer-segmentation engines (RFM, K-Means, GMM, DBSCAN) — 6 optimal clusters, 89% month-over-month stability, +23% campaign response rate.

03 Selected Systems

Verxlite

2026 · V0.1.0

[Universal AI workflow agent](#) · [github.com/gadda00/verxlite](#)

Multi-tenant SaaS automating post-meeting followups, lead assignment, support triage and approval routing across Gmail, Calendar and HubSpot. FastAPI + Celery + Next.js 14, JWT + Clerk webhooks, Google/HubSpot OAuth, Fernet-encrypted tokens, Langfuse observability — shipped with a transparent 281-defect code review and 83 backend tests passing.

Fraud Detection System

2021 — PRESENT · V2.0

Real-time scoring engine · github.com/gadda00/fraud-detection-system

Enterprise-grade Go engine: **7 statistical detectors** (Z-Score, IQR, Velocity, Geo-Distance, Device Fingerprint, Merchant Risk, Behavioral Anomaly) fused by a weighted-vote ensemble, logistic-regression calibrated, with a deterministic rules overlay. **~33 µs/score, 84% recall, 1.37% FPR**. Case management, RBAC (API key + JWT), Prometheus + OpenTelemetry, Slack alerts. PCI-DSS / PSD2 / GDPR Article 22 ready. Distroless image (~15 MB) on Kubernetes with HPA.

Financial Forecasting Suite

2020

Time-series toolkit · github.com/gadda00/financial-forecasting-suite

Holt-Winters, ARIMA/SARIMA, Prophet-style decomposition and ensemble forecasting with automatic model selection — **87.3% mean accuracy (1-MAPE)** across 12 metrics with 12–24 month horizons.

04 Technical Skills

LANGUAGES	Python, Go, TypeScript / JavaScript, Dart, SQL, Bash
LLMS	GPT, Claude, Gemini, GLM-4.6, Kimi K2, MiMo, Mistral, Qwen, DeepSeek, Llama
AGENT SYSTEMS	LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, custom multi-agent DAG orchestration
RAG & RETRIEVAL	pgvector, Pinecone, Weaviate, hybrid search (BM25 + vector), reranking, chunking strategies
ML & STATISTICS	TensorFlow / TFLite, scikit-learn, Holt-Winters, GARCH/VaR, K-Means++, OLS, Granger causality, Isolation Forest, weighted-vote ensembles
WEB & MOBILE	Next.js, React, Node.js, FastAPI, Gin (Go), Flutter, Celery
DATA & INFRA	PostgreSQL, Redis, Kafka, Docker, Kubernetes, distroless containers
CLOUD & OPS	Google Cloud (Earth Engine), CI/CD, Git/GitHub, Prometheus, OpenTelemetry, Grafana
ALSO	Fraud detection & risk scoring, OAuth (Google, HubSpot), JWT auth, LLM-as-judge, tracing, evals & observability

05 Education & Certifications

B.Sc. Computer Science

2014 — 2018

University of Embu · Embu, Kenya

- Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Statistics, Software Engineering, Distributed Systems.
- Final-year project: an early crop advisory system that seeded the KilimoPRO concept. Top-quartile standing in the School of Computing.

AI Engineer for Data Scientists Associate

IN PROGRESS

Data management, exploratory analysis, governance and modern application development for AI.

AI Engineer for Developers Associate

IN PROGRESS

Building and deploying AI-powered applications — programming, model development, responsible AI.

06 Writing & Languages

- Author of **10 long-form technical articles** (22,000+ words) on multi-agent DAG architecture, real math vs vibes, multi-LLM councils, quant finance for agriculture, and the AgentReplay whitepaper — victorndunda.com/blog.
- Author of **8 practical AI guides** for SMBs — ChatGPT for business, prompt engineering, no-code chatbots, automation, AI for farmers, Swahili-first AI, RAG in one day.

English · Professional

Swahili · Native

French · Basic